**BRAINWARE UNIVERSITY**
**School of Engineering**
**Department of Computer Science & Engineering – Cyber Security & Data Science**
*398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125*

# Review-I (A.Y.-2026-27)

Project-I, PROJ-CSD782 | Program: B.Tech. CSE-DS [Batch: 2023]

| S.N. | Roll No. | Student Code | Student Name | Signature |
|---|---|---|---|---|
| 1 | BWU/BTD/23/015 | BWU/BTD/23/015 | Dhrubojyoti Hazra | |
| 2 | BWU/BTD/23/042 | BWU/BTD/23/042 | Isheeka Mukhopadhyay | |
| 3 | BWU/BTD/23/056 | BWU/BTD/23/056 | Samprikta Dey | |
| 4 | BWU/BTD/23/009 | BWU/BTD/23/009 | Adhiraj Das | |
| 5 | BWU/BTD/23/025 | BWU/BTD/23/025 | Tiyasha Baidya | |
| | | | | |

Project Title: **Fraud Detection in Digital Payment Systems Using Machine Learning**

*Build a fraud detection framework capable of analyzing large volumes of digital transaction data stored in a data warehouse. Machine learning and anomaly detection techniques identify suspicious transactions in real time. The system generates risk scores and alerts to help financial institutions prevent fraudulent activities.*

Name & Signature of Project Supervisor
Rudranath Mitra

Dr. Ranadip Kundu
Project Coordinator CSE – CS & DS

**Project Progress (up to Review 1):**

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**1. Problem Domain and Its Analysis**

India's shift to UPI and card-based digital payments has been matched by increasingly sophisticated, fast-evolving fraud tactics. FinTech platforms face a direct trade-off: undetected fraud causes financial loss, while over-aggressive detection generates false positives that block genuine customers. This precision–recall trade-off is the core problem addressed by this project.

**2. Study of Existing Systems and Feasibility**

Traditional rule-based fraud systems rely on fixed thresholds and are widely reported to fail against novel, fast-evolving fraud patterns, since rules demand constant manual updates. Machine-learning approaches improve accuracy but carry well-documented limitations of their own: severe class imbalance (fraud typically under 1% of transactions), concept drift, and — most persistently — limited interpretability that restricts real-world trust and adoption. Commercial systems confirm this gap: Stripe Radar added a separate "Risk Insights" layer purely to explain its ML decisions to merchants, while Razorpay's Thirdwatch and Feedzai market explainable, hybrid rule-plus-ML scoring as a direct response to black-box alert fatigue. These systems are, however, proprietary and built on private transaction data, making them unavailable for direct study or replication.

Feasibility for this project rests on three points: the Kaggle Credit Card Fraud dataset (284,807 transactions, 492 confirmed frauds, 0.17% fraud rate) is the same public benchmark used across current fraud-detection literature, providing a validated substitute for proprietary data; a managed backend (Supabase) removes most infrastructure work, focusing semester effort on modeling and workflow design; and the required stack (Python/scikit-learn, FastAPI, Docker) matches the team's existing skillset.

**3. Objectives, Methodology, Planning & Significance**

**Objectives:** build a dual-scoring engine (XGBoost for known fraud patterns, Isolation Forest for zero-day anomalies); handle class imbalance via SMOTE and class weighting; attach a SHAP/LIME explainability layer so every flagged transaction carries a stated reason; implement real-time scoring (FastAPI + Redis) fed by both batch uploads and a live transaction simulator; build a role-based dashboard (Analyst / Senior Analyst / Team Lead) spanning a simulated multi-branch environment with Postgres Row-Level-Security-enforced access, and an automated critical-case escalation workflow with SLA tracking.

**Methodology:** preprocessing and feature engineering → model training and evaluation (F1, Precision, Recall, AUC-ROC, False Positive Rate on a 20% holdout) → integration with a parallel rules engine → validation against scenario-based test cases (velocity spikes, geographic anomalies, zero-day patterns) → refinement from logged analyst feedback.

**Planning:** work is split across five members (ML pipeline, backend/scoring service, dashboard/explainability, data/deployment) over a phased build spanning the semester, from schema and ingestion through to reliability hardening and final demonstration.

**Significance:** the system directly targets the interpretability, adaptability, and human-oversight gaps identified above, translating a validated real-world problem into a small-scale but architecturally sound fraud-detection platform.